

Hatchery Documentation (Olariu Dan-Vasile)

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2. Subsystem Use cases

A. UC-01: Create the Egg Reception

Use Case ID: UC-01

Use Case Name: Create the Egg Reception

Actor: Logistics Manager

Preconditions:

- External breeder farm delivery notice received
- Eggs physically delivered to hatchery
- Logistics Manager is authenticated

Main Flow:

1. System receives Egg Delivery Notice from external breeder farm
2. auto-creates a draft **EggReception** record
3. Logistics Manager reviews incoming delivery details (farm name, date, quantity)
4. Manager selects storage location and responsible worker.
5. Manager inputs additional details:
 - Transport temperature and humidity
 - Arrival conditions
 - Truck information
6. Manager confirms reception
7. System finalizes EggReception record and creates associated EggBatch
8. System updates stock and sets eggs as Awaiting Inspection

Alternative Flows:

- **A1: Missing data**
 - System displays warning and requests correction
- **A2: Duplicate batch**
 - System blocks and displays existing batch ID

Postconditions:

- New EggReception and EggBatch created
- Eggs marked as received and pending quality inspectio

Business Rules:

- One reception per delivery notice
 - Reception must be approved within 24 hours
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B. UC-02: Create Egg Quality Report

Use Case ID: UC-02

Use Case Name: Create Egg Quality Report

Actor: Worker

Preconditions:

- EggReception exists with status = “Awaiting Inspection”
- Worker is authenticated

Main Flow:

1. Worker accesses Egg Quality Report form
2. System retrieves data from related *EggReception*
3. Worker inspects eggs and records:
 - Total inspected
 - Cracked eggs
 - Dirty eggs
 - Unsuitable eggs
 - Remarks
4. System calculates acceptance percentage automatically
5. Worker submits report for approval
6. Hatchery Manager reviews and approves the report
7. System updates EggBatch status to “Ready for Incubation.”

Alternative Flows:

- **A1: Missing inspection values**
 - System requests correction
- **A2: Reject batch**
 - System marks as “Rejected” and notifies Logistics Manager

Postconditions:

- EggQualityReport created and linked to EggReception
- Batch status updated accordingly

Business Rules:

- Inspection required for every reception
- Rejected batches cannot be incubated

C. UC-03: Create Incubation Sheet

Use Case ID: UC-03

Use Case Name: Create Incubation Sheet

Actor: Incubation Technician

Preconditions:

- EggBatch exists and is approved for incubation
- Incubator available and operational

Main Flow:

1. Technician accesses Incubation Module
2. System lists available incubators and approved egg batches
3. Technician selects incubator and assigns batch
4. System auto-creates IncubationCycle record
5. Technician sets incubation start date, temperature, humidity, and turning frequency
6. System generates Incubation Sheet linked to the incubator
7. Technician confirms and system records incubation start timestamp

Alternative Flows:

- **A1: No Incubator Available**
 - System displays “All incubators busy”
- **A2: Invalid settings**
 - System requests correction

Postconditions:

- IncubationSheet created and linked to Incubator and EggBatch
- Cycle begins tracking automatically

Business Rules:

- One batch per incubator cycle

- Temperature and humidity must be within defined operational limit

D. UC-04: Update Incubation Sheet

Use Case ID: UC-04

Use Case Name: Update Incubation Sheet

Actor: Incubation Technician

Preconditions:

- IncubationCycle in progress

Technician is authenticatedMain Flow:

1. Technician opens active Incubation Sheet
2. System displays current readings for temperature, humidity, and turning
3. Technician adds daily entries:
 - Date/time
 - Temperature
 - Humidity
 - Notes (if any anomalies)
4. System validates data and updates IncubationSheetLine
5. Technician can edit or delete entries if errors found
6. System recalculates daily averages and saves

Alternative Flows:

- A1: Data missing
 - System highlights incomplete fields
- A2: Out-of-range values
 - System shows warning

Postconditions:

- Updated incubation records stored
- Historical tracking of incubation performance maintained

Business Rules:

- One incubation per cycle
- Technicians can modify data within 24 hours
- After 24 hours it requires manager approval

E. UC-05: Create Chick Delivery Notice

Use Case ID: UC-05

Use Case Name: Create Chick Delivery Notice

Actor: Logistics Manager

Preconditions:

- IncubationCycle completed successfully
- Chicks counted and separated
- Logistics Manager is authenticated

Main Flow:

1. System Logistics Manager accesses Chick Delivery module
2. System retrieves completed batches (status = "Hatched").
3. Manager creates a new ChickDeliveryNotice
4. Inputs data:
 - Batch ID
 - Destination: Broiler Farm
 - Total chicks
 - Average weight
 - Transport details
5. System validates quantities and vehicle capacity
6. Manager confirms and submits for approval
7. System updates status to "Approved" and "Ready for Dispatch"

Alternative Flows:

- **A1: Insufficient chicks**
 - System requests manual adjustment
- **A2: Invalid destination**
 - System rejects entry

Postconditions:

- Chick Delivery Notice created and linked to batch
- Batch status updated to "Dispatched"

Business Rules:

- Delivery notice must be sent before physical dispatch
 - Only approved lots can be shipped
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F. UC-06: Create Vaccination and Treatment Sheet

Use Case ID: UC-06

Use Case Name: Create Vaccination and Treatment Sheet

Actor: Veterinarian

Preconditions:

- Hatch event completed
- Chicks in post-hatch holding area

Main Flow:

1. Veterinarian opens Vaccination/Treatment Sheet form
2. System loads hatched batches awaiting treatment
3. Vet selects batch and records:
 - Vaccine
 - Medication
 - Dosage
 - Administration method
 - Date
 - Time
4. System validates availability in stock
5. Vet confirms and submits
6. System updates treatment history for the batch

Alternative Flows:

- **A1: Out of stock**
 - System prevents submission and notifies Logistics Manager
- **A2: Duplicate entry**
 - System rejects duplicate treatments

Postconditions:

- Vaccination/Treatment Sheet created and approved
- Treatment data linked to batch record

Business Rules:

- Only authorized vets can create treatment sheets
 - One sheet per batch per treatment session
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G. UC-07: Creating the Daily Sanitation Report**Use Case ID:** UC-07**Use Case Name:** Creating the Daily Sanitation Report**Actor:** Worker**Preconditions:**

- Worker assigned to sanitation tasks
- Hatchery area active for operations

Main Flow:

1. **[Time Event: Daily - Every 24 hours]** System prompts the Worker to create a new Daily Sanitation Report
2. Worker opens the sanitation report form
3. System pre-fills date and area details based on daily schedule
4. Worker records the following:
 - Area cleaned
 - Type of disinfectant used
 - Duration of cleaning
 - Supervisor name
5. Worker submits the report for review
6. Hatchery Manager reviews and approves the report
7. System marks the report as “Approved” and archives it for traceability

Alternative Flows:

- **A1: Worker unavailable**
 - System reassigns the task to another user in the same role

Postconditions:

- Daily Sanitation Report successfully created, approved, and linked to the hatchery’s sanitation log

Business Rules:

- One Sanitation Report per area per day

- Reports must be approved before the next daily cycle starts
- Historical data must remain immutable after approval
- Withdrawal period must be respected
- Cannot delete active treatment (can only deactivate)
- Medication batch numbers must be recorded

Extends:

- Create the Consumption Sheet (auto-creates medication consumption entries)
- Update the Consumption Sheet (updates daily with treatment dosages)

H. UC-08: Generate Weekly Production Summary

Use Case ID: UC-08

Use Case Name: Generate Weekly Production

Actor: Hatchery Manager

Preconditions:

- All EggReceptions, IncubationSheets, VaccinationSheets, and Chick Deliveries for the week are completed
- Hatchery Manager is authenticated

Main Flow:

1. **[Time Event: Weekly – Every 7 days]** System automatically generates a draft Weekly Production Summary
2. System aggregates data from all relevant modules:
 - Egg Receptions (total eggs received)
 - Incubation Sheets (temperature & humidity averages)
 - Hatching Batches (fertility & hatch rate)
 - Chick Deliveries (quantity and average weight)
3. System calculates key performance indicators (KPI):
 - Hatch Rate (%)
 - Fertility Rate (%)
 - Total Chicks Delivered
 - Incubator Occupancy Rate
4. Hatchery Manager reviews and validates the generated summary

5. Manager may add remarks or corrections
6. System finalizes the report and marks it as “Approved”
7. Weekly Production Summary stored in the database and made available for management dashboards

Alternative Flows:

- **A1: Missing Data**
 - System flags the summary as “Partial” and lists the missing sources

Postconditions:

- Weekly Production Summary created, validated, and archived
- KPIs visible to authorized roles (Hatchery Manager, Vet, Logistics Manager)

Business Rules:

- One summary per week per hatchery
- Automatic generation occurs each Monday at 00:00
- Only Managers can approve or modify summaries

3. Entities and the logic of them

Entity Definitions, relationships and business limitations

a) BaseEntity (Abstract)

- id: Long (PK)
- createdAt: DateTime
- createdBy: Employee
- updatedAt: DateTime
- updatedBy: Employee
- isActive: Boolean
- version: Integer (for optimistic locking)

b) Employee

- employeeId: Long (PK)
- firstName: String
- lastName: String
- email: String
- phone: String
- role: Enum (HATCHERY_MANAGER, LOGISTICS_MANAGER, INCUBATION_TECHNICIAN, WORKER, VETERINARIAN)
- hireDate: Date
- isActive: Boolean
- hatchery: Hatchery (FK)

c) Hatchery

- hatcheryId: Long (PK)
- location: String
- address: String
- capacity: Integer (total eggs)
- incubatorCount: Integer
- manager: Employee (FK)

Relationships:

- Has many Incubators, EggReceptions, and HatchingBatches

d) Incubator

- incubatorId: Long (PK)
- code: String
- capacity: Integer
- status: Enum (AVAILABLE, IN_USE, MAINTENANCE)
- hatchery: Hatchery (FK)

Business rules:

- One batch active per incubator

e) EggBatch

- batchId: Long (PK)
- sourceFarm: String
- receptionDate: Date
- quantityReceived: Integer
- acceptedQuantity: Integer
- status: Enum (PENDING_INSPECTION, APPROVED, REJECTED, INCUBATED, HATCHED)

f) EggReception

- receptionId: Long (PK)
- date: Date
- sourceFarm: String
- transportTemperature: Decimal
- responsibleEmployee: Employee (FK)
- totalQuantity: Integer

Relationships:

- Has many EggReceptionLines

Business Rules:

- Created from external delivery notice

g) EggReceptionLine

- lineId: Long (PK)
- reception: EggReception (FK)
- trayNumber: Integer
- eggsCount: Integer
- remarks: String

h) EggInspection

- inspectionId: Long (PK)
- eggBatch: EggBatch (FK)
- cracked: Integer
- dirty: Integer
- unsuitable: Integer
- remarks: String

Business rules:

- Inspection mandatory before incubation

i) IncubationCycle

- cycleId: Long (PK)
- incubator: Incubator (FK)
- eggBatch: EggBatch (FK)
- startDate: Date
- endDate: Date
- status: Enum (IN_PROGRESS, COMPLETED, FAILED)

j) IncubationSheet

- sheetId: Long (PK)
- incubationCycle: IncubationCycle (FK)
- technician: Employee (FK)
- startDate: Date

Relationships:

- Has many IncubationSheetLines

k) IncubationSheetLine

- lineId: Long (PK)
- incubationSheet: IncubationSheet (FK)
- dateTime: DateTime
- temperature: Decimal
- humidity: Decimal
- notes: String

l) VaccinationSheet

- sheetId: Long (PK)
- batch: EggBatch (FK)
- veterinarian: Employee (FK)
- date: Date
- vaccineName: String
- dosage: Decimal

Relationships:

- Has many VaccinationSheetLines

m) VaccinationSheetLine

- lineId: Long (PK)
- vaccinationSheet: VaccinationSheet (FK)
- administrationMethod: Enum (SPRAY, INJECTION, DRINKING_WATER)
- quantityUsed: Decimal
- remarks: String

n) ChickDeliveryNotice

- noticeId: Long (PK)
- batch: EggBatch (FK)
- destinationFarm: String
- transportManager: Employee (FK)
- vehicleInfo: String
- totalChicks: Integer

- averageWeight: Decimal
- status: Enum (DRAFT, APPROVED, SENT)
 - o) ChickDeliveryNotice
- lineId: Long (PK)
- chickDeliveryNotice: ChickDeliveryNotice (FK)
- crateNumber: Integer
- quantity: Integer
- remarks: String
 - p) SanitationReport
- lineId: Long (PK)
- date: Date
- area: String
- worker: Employee (FK)
- status: Enum (DRAFT, APPROVED)
 - q) SanitationReportLine
- lineId: Long (PK)
- sanitationReport: SanitationReport (FK)
- disinfectantUsed: String
- duration: Integer
- verifiedBy: Employee (FK)
 - r) ProductionSummary
- summaryId: Long (PK)
- weekNumber: Integer
- hatchery: Hatchery (FK)
- totalEggsReceived: Integer
- totalChicksHatched: Integer
- hatchRate: Decimal
- reportDate: Date